

Data Preprocessing

After completing the data exploration phase, the next critical step in a data science project is data preprocessing. This stage involves preparing your tabular data for analysis or machine learning modeling by cleaning and transforming it. Here's a step-by-step guide to perform data preprocessing.

Note: You have to **split the data** into training and test data before applying this data preprocessing.

1. Handling Missing Values

- **Explanation:** Missing data can distort analyses if not properly handled. Depending on the context, fill missing values using techniques like mean, median, mode imputation, or use more complex methods like k-NN imputation. Alternatively, rows or columns with a high percentage of missing values might be dropped.

2. Handling Outliers

- **Explanation:** Outliers can significantly impact model performance. Identify outliers using IQR (Interquartile Range) or Z-score methods and then either remove them or cap them using appropriate percentile values.

3. Feature Scaling

- **Explanation:** Different numerical scales across features can lead to biased models. Normalize or standardize features to bring them to a similar scale. Common methods include Min-Max Scaling, Z-score Standardization, and Robust Scaling.

4. Feature Engineering

- **Explanation:** Create new features from existing ones to improve model performance. This can involve creating interaction terms, binning variables, or extracting components from date/time variables.

5. Data Transformation

- **Explanation:** Apply transformations to make the data more suitable for analysis. Common transformations include log transformation, square root transformation, or power transformation to handle skewed data.

6. Dimensionality Reduction

- **Explanation:** High-dimensional datasets can lead to overfitting and long training times. Use techniques like PCA (Principal Component Analysis) or feature selection methods to reduce the number of variables while retaining important information.

7. Dealing with Imbalanced Data

- **Explanation:** Imbalanced datasets can lead to poor model performance, particularly for the minority class. Techniques like SMOTE (Synthetic Minority Over-sampling Technique) or undersampling can help balance the classes.

8. Encoding Categorical Variables

- **Explanation:** Machine learning models typically require numerical input, so convert categorical variables using methods like one-hot encoding (for nominal variables) or ordinal encoding (for ordinal variables).

Documentation

- **Explanation:** Document every preprocessing step taken, including the rationale behind each decision. This documentation is crucial for reproducibility and understanding the impact of preprocessing on the outcomes.